

Measuring AI Ability to Complete Long Software Tasks

Time Horizon Doubling Every 7 Months

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Source: <https://arxiv.org/abs/2503.14499>

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A New Metric: 50% Task-Completion Time Horizon

Methodology Flow:

1. Create diverse task suite of 170 tasks.
2. Measure human timing and AI success rates.
3. Fit logistic model to find the task duration at which AI succeeds 50% of the time.

Family	Length	Description
find_shell_script	3 seconds	Multiple choice: “Which file is a shell script?” Choices: “run.sh”, “run.txt”, “run.py”, “run.md”
wikipedia_research	1 minute	Research simple factual information from Wikipedia and provide accurate answers to straightforward questions.
oxdna_simple	9 minutes	Detect and fix a bug in the input files for a molecular dynamics simulation using the oxDNA package.
munge_data	56 minutes	Write a Python script to transform JSON data from one format to another by inferring the conversion rules from provided example files.
cuda_backtesting	8 hours	Speed up a Python backtesting tool for trade executions by implementing custom CUDA kernels while preserving all functionality, aiming for a 30x performance improvement.

170 Tasks Spanning 3 Seconds to 8 Hours of Human Effort

Suite Composition: HCAST (97 tasks), RE-Bench (7 tasks), SWAA (66 tasks)

Task Family	Human Time	Description
find_shell_script	3 seconds	Multiple choice: "Which file is a shell script?"
wikipedia_research	1 minute	Research simple factual information from Wikipedia
oxdna_simple	9 minutes	Detect and fix a bug in molecular dynamics simulation input files
munge_data	56 minutes	Write a Python script to transform JSON data by inferring conversion rules
cuda_backtesting	8 hours	Implement custom CUDA kernels for 30x speedup of a backtesting tool

AI Task Horizon Grew from 2 Sec to ~1h 50m in 6 Years

Exponential trend: 50% time horizon
vs. model release date.

Doubling time: ~207 days (~7 months).

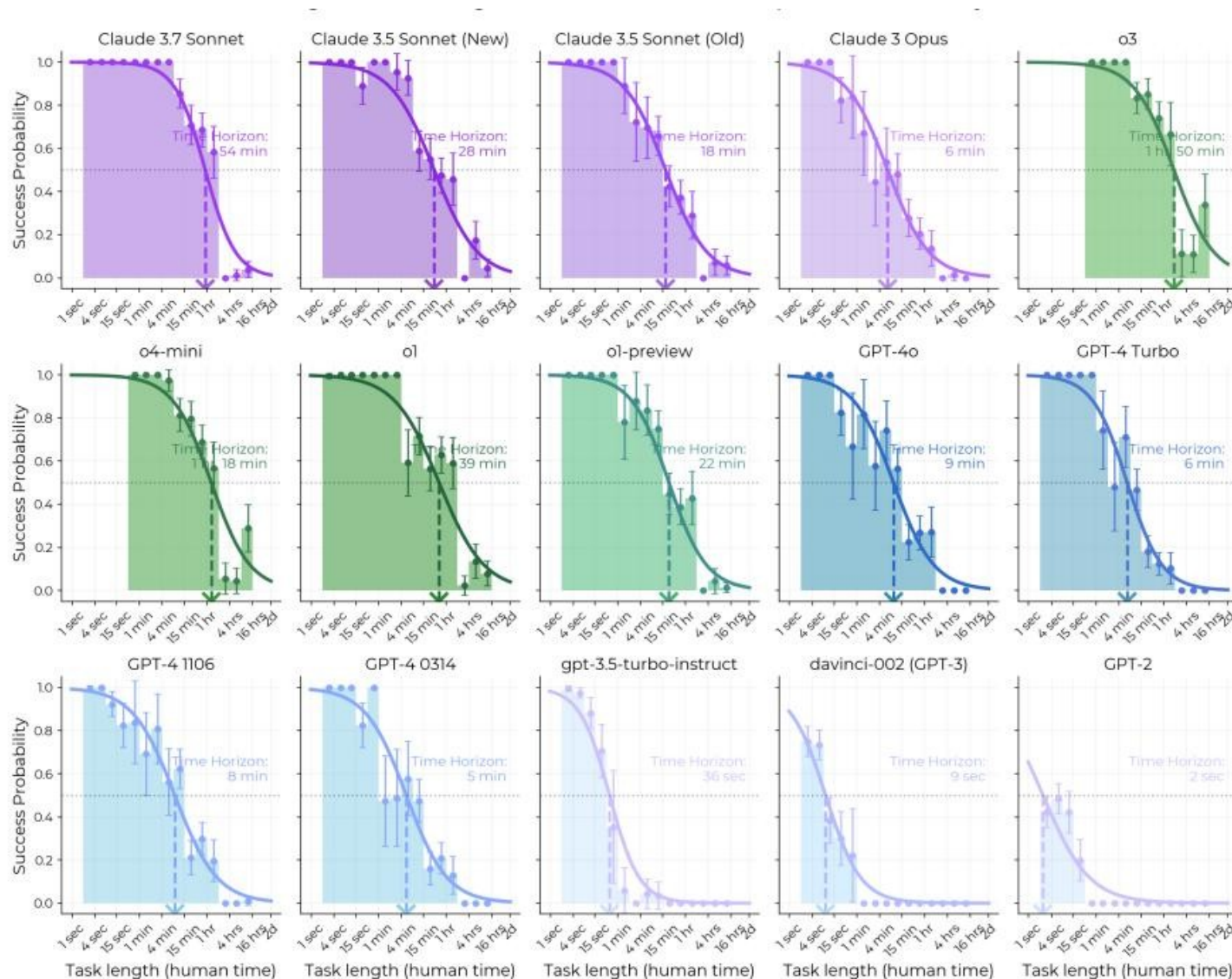
Extremely strong fit: $R^2 = 0.97$.

GPT-2 (2019): ~2 seconds

GPT-4 0314 (2023): ~5 minutes

Claude 3.7 Sonnet (2025): ~54 minutes

o3 (2025): ~1 hour 50 minutes



Success Probability Drops Sharply as Task Length Increases

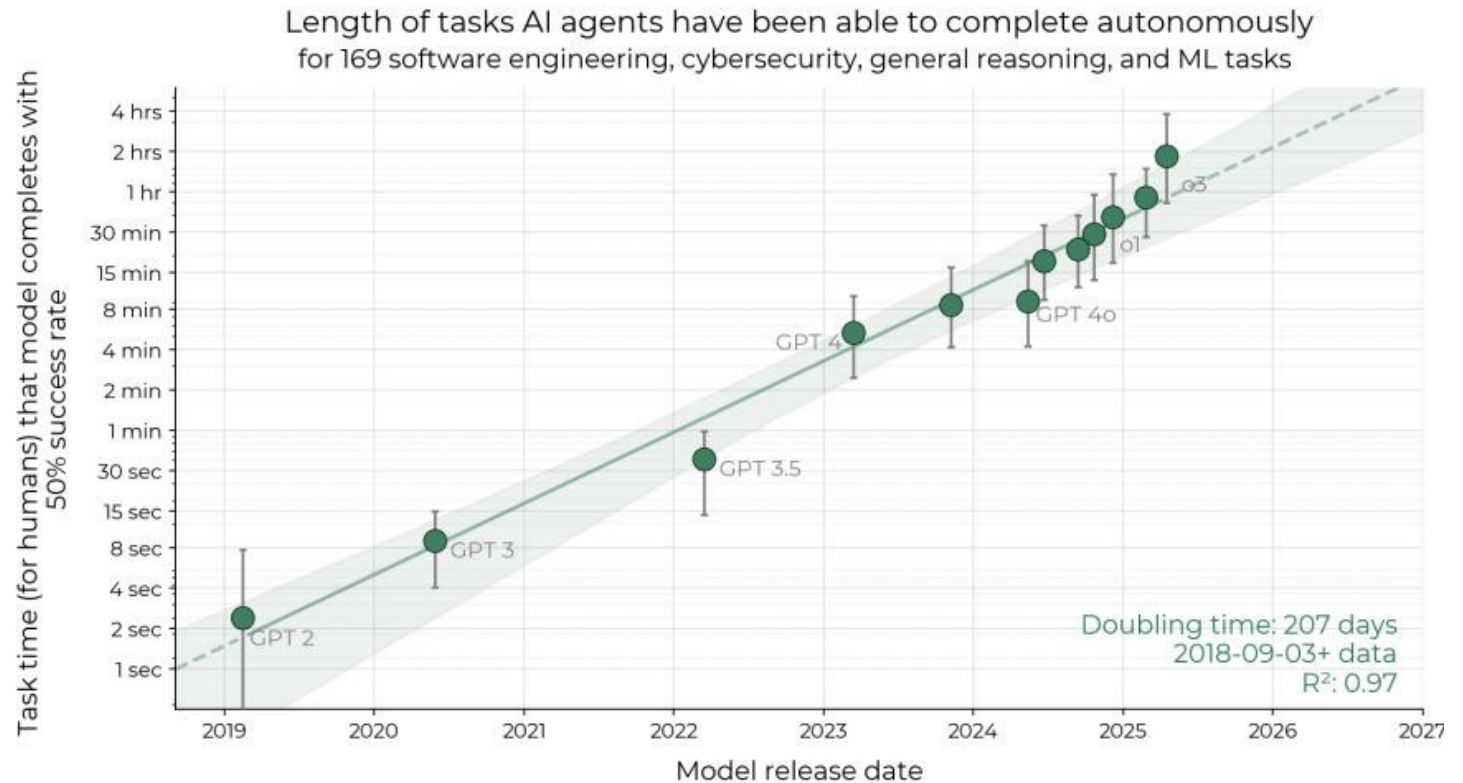
Per-model success probability curves show decline with task length.

Frontier models push the curve significantly rightward.

GPT-2: steep drop almost immediately (2 sec 50% horizon).

Claude 3.7 Sonnet maintains 50% success up to 54 minutes.

o3 extends the 50% success threshold to 1 hour 50 minutes.



Reasoning, Tool Use, and Error Recovery Drive Growth

- Improved logical reasoning and systematic problem-solving.
- Better tool use (e.g., executing scripts, analyzing outputs).
- Greater reliability and self-awareness in long-horizon execution.
- Enhanced ability to adapt to mistakes and iteratively correct them.
- Key Limitation: Performance remains notably lower on unstructured or 'messier' tasks.

Extrapolation Predicts Month-Long Task Automation by 2028-2031

- Naïve forward extrapolation of the current trend predicts profound capabilities.
- Target: 1-month time horizon (approx. 167 work hours).
- Estimated timeline: Reaching this milestone between mid-2028 and mid-2031.
- Acceleration note: The 2023-2025 growth rate is ~20% faster than the overall 2019-2025 rate.
- Caveats: External validity remains an open question. Tasks may not perfectly represent the average segment of real-world intellectual labor.

Takeaways: Exponential Growth Demands Proactive Governance

1. Metric Robustness: The 50% time horizon is intuitive and reliable ($R^2=0.97$ over 6 years).
2. Growth Rate: Doubling every ~7 months, with potential acceleration since 2024.
3. Current Frontier: Top agents (e.g., o3) successfully complete tasks taking humans ~2 hours.
4. Real-World Translation: External validity is uncertain; benchmark tasks may not fully capture the complexity of all real-world jobs.
5. Safety Implications: Rapid trajectory towards month-long autonomous capabilities necessitates proactive safety governance and workforce planning.